

# MOCKUP — Why Preparation-Only and Cross-Stage Differ at the GPU

## Illustrative service units; non-preemptive request execution

### Workload

Both Tenant A and Tenant B are continuously backlogged.  
Every request has the same CPU preparation cost: 1 worker-time unit.

**Tenant A request** GPU cost = 4 units (long inference)  
**Tenant B request** GPU cost = 1 unit (short inference)

### Preparation-only max-min

CPU received service: A = 8 units, B = 8 units

CPU preparation



time →

GPU inference



time →

**GPU received service: A = 12 units, B = 3 units → gap = 9**

After fair preparation, native/FCFS inference admits equal request counts—not equal GPU service.

### Cross-stage max-min

CPU received service: A = 8 units, B = 8 units

CPU preparation



time →

GPU inference



time →

**GPU received service: A = 8 units, B = 8 units → gap = 0**

Inference admission chooses the least-served tenant; B receives more requests because each is cheaper.